Sum of a random number of independent and identically distributed random variables

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Sum of a random number of i.i.d. random variables

Let N and $X_1, X_2, \ldots, X_k, \ldots$ be independent random variables such that:

- N takes nonnegative integer values.
- The variables X_k are identically distributed.

We consider the sum

$$S = X_1 + X_2 + \cdots + X_N,$$

where the number of terms is itself random and given by N. (We define $S = 0$ when $N = 0$.)

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Conditioning on the possible values of N allows the characteristic function of the random variable S to be expressed as follows:

$$\begin{aligned} M_S(\omega) &= \mathbb{E}\left(e^{i\omega S}\right) \\ &= \sum_{n \geqslant 0} \mathbb{E}\left(e^{i\omega S} \mid N = n\right) \mathbb{P}(N = n) \end{aligned}$$

- Observe that if $N = 0$, then $\mathbb{E}\left(e^{i\omega S} \mid N = 0\right) = \mathbb{E}\left(e^{i\omega 0}\right) = 1$.

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If $n \geqslant 1$, then

$$\begin{aligned}
\mathbb{E}\left(e^{i\omega S} \mid N = n\right) &= \mathbb{E}\left(e^{i\omega(X_1+\cdots+X_N)} \mid N = n\right) \\
&= \mathbb{E}\left(e^{i\omega(X_1+\cdots+X_n)} \mid N = n\right) \\
&= \mathbb{E}\left(e^{i\omega(X_1+\cdots+X_n)}\right) \\
&= \mathbb{E}\left(e^{i\omega X_1}\right) \cdots \mathbb{E}\left(e^{i\omega X_n}\right) \\
&= M_X(\omega)^n
\end{aligned}$$

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Recall that the probability generating function of N is defined as $G_N(z) = \sum_{n \geqslant 0} z^n \, \mathbb{P}(N = n)$. (Here we consider $G_N(z)$ as a function of a complex variable z.)

Proposition

The characteristic function of the random sum S is given by:

$$M_S(\omega) = \sum_{n \geqslant 0} M_X(\omega)^n \, \mathbb{P}(N = n) = G_N(M_X(\omega))$$

- The composition $G_N(M_X(\omega))$ is well-defined, because $|M_X(\omega)| \leqslant 1$ for all $\omega \in \mathbb{R}$ and $G_N(z)$ converges for all $z \in \mathbb{C}$ such that $|z| \leqslant 1$.

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Remark. The preceding argument may be summarised as follows:

$$M_S(\omega) = \mathbb{E}\left(e^{i\omega S}\right) = \mathbb{E}\left(\mathbb{E}\left(e^{i\omega S} \mid N\right)\right)$$

Now observe that $\mathbb{E}\left(e^{i\omega S} \mid N\right)$ is the random variable $M_X(\omega)^N$,
because $\mathbb{E}\left(e^{i\omega S} \mid N = n\right) = M_X(\omega)^n$,

Therefore, since $G_N(z) = \mathbb{E}(z^N)$,

$$M_S(\omega) = \mathbb{E}\left(M_X(\omega)^N\right) = G_N(M_X(\omega))$$

Example 1

- Let N represent the number of customers arriving at a service point, modelled as a $\text{Po}(\lambda)$-distributed random variable.
- Assume that each arrival is independently served with probability p.

  (Therefore, an arrival is not served with probability $q = 1 - p$.)

We now determine the probability distribution of the number of customers served, denoted by S.

Example 1

We have

$$S = X_1 + \cdots + X_N,$$

where X_k is the indicator random variable of the event "the k-th arrival has been served". So, $X_k \sim \text{Be}(p)$.

Moreover,

$$M_X(\omega) = q + pe^{i\omega}, \quad \omega \in \mathbb{R}$$

$$G_N(z) = e^{\lambda(z-1)}, \quad z \in \mathbb{C}$$

Example 1

Therefore

$$\begin{aligned} M_S(\omega) &= G_N\left(M_X(\omega)\right) = G_N(q + pe^{i\omega}) \\ &= e^{\lambda(z-1)}\Big|_{z=q+pe^{i\omega}} = e^{\lambda(q+pe^{i\omega}-1)} \\ &= e^{\lambda p(e^{i\omega}-1)} \end{aligned}$$

This is the characteristic function of a Poisson random variable with parameter λp. Hence,

$$S \sim \mathrm{Po}(\lambda p)$$

Example 1

Analogously, if R denotes the number of non-served customers, then $R \sim \text{Po}(\lambda q)$.

- It can be proved that S and R are independent variables.
- Notice that the convolution theorem holds:

$$
\begin{aligned}
M_S(\omega)\, M_R(\omega) &= e^{\lambda p(e^{i\omega}-1)}\, e^{\lambda q(e^{i\omega}-1)} \\
&= e^{\lambda(p+q)(e^{i\omega}-1)} = e^{\lambda(e^{i\omega}-1)} \\
&= M_N(\omega),
\end{aligned}
$$

in accordance with the fact that $S + R = N$.

Example 2

As a second example consider the following scenario.

- The number N of customers arriving at a service point is a $\mathrm{Ge}(p)$-distributed random variable.
- The service times $X_k,\ k \geqslant 1$, are independent and $\mathrm{Exp}(\mu)$-distributed random variables.

Let S represent the total occupancy time of the service point,

$$S = X_1 + \cdots + X_N$$

Example 2

Now

$$M_X(\omega) = \frac{\mu}{\mu - i\omega}, \quad G_N(z) = \frac{pz}{1 - qz}$$

Therefore

$$M_S(\omega) = G_N\left(M_X(\omega)\right) = G_N\left(\frac{\mu}{\mu - i\omega}\right)$$

$$= \left.\frac{pz}{1 - qz}\right|_{z=\frac{\mu}{\mu - i\omega}} = \frac{p\,\frac{\mu}{\mu - i\omega}}{1 - q\,\frac{\mu}{\mu - i\omega}}$$

$$= \frac{\mu p}{\mu p - i\omega}$$

We conclude that

$$S \sim \mathrm{Exp}(\mu p)$$

Expected values

If we focus exclusively on the expectation of the random sum $S = X_1 + \cdots + X_N$, we have

$$\mathbb{E}(S) = \mathbb{E}\left(\mathbb{E}(S \mid N)\right)$$

Let m denote the common expected value of the variables X_k. If $N = n$, then

$$\begin{aligned} \mathbb{E}(S \mid N = n) &= \mathbb{E}(X_1 + \cdots + X_N \mid N = n) \\ &= \mathbb{E}(X_1 + \cdots + X_n) \\ &= \mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n) = n \cdot m, \end{aligned}$$

Therefore $\mathbb{E}(S \mid N)$ is the random variable mN and so

$$E(S) = \mathbb{E}(mN) = m\,\mathbb{E}(N) = \mathbb{E}(N)\,\mathbb{E}(X).$$

Expected values

- For instance, in Example 1 we had $X \sim \mathrm{Be}(p)$, $N \sim \mathrm{Po}(\lambda)$, and $S \sim \mathrm{Po}(\lambda p)$. Consequently,

$$\mathbb{E}(S) = \lambda p = \mathbb{E}(N)\,\mathbb{E}(X)$$

- Analogously, in Example 2, $X \sim \mathrm{Exp}(\mu)$, $N \sim \mathrm{Ge}(p)$, and $S \sim \mathrm{Exp}(\mu p)$. Hence

$$\mathbb{E}(S) = \frac{1}{\mu p} = \frac{1}{p} \cdot \frac{1}{\mu} = \mathbb{E}(N)\,\mathbb{E}(X)$$